

Learning from Its Own Footprint

Strategic Disclosure, Capital Feedback, and the Design of AI Valuation Rules in IPO Allocation

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KEY MESSAGE In the model, outcomes can be high partly BECAUSE the rule priced issuers highly. Naive history learns $b \approx 6.36$ vs. true 1.02 (feedback bias 5.344) and raises mean manipulation $0.408 \rightarrow 1.367$ (+235%) under uniform-price allocation. Capital-purged history reduces bias, but does not remove it.

01 Question & model

Research question & verified gap

How should an AI valuation rule be calibrated when the issuer manages its input and the rule's own pricing changes what it later learns from?
Application setting: AI-assisted IPO/equity valuation with auction-based capital allocation.
Gap: costly-disclosure models hold the rule fixed; IPO work treats signals as exogenous; performativity work does not jointly model allocation and later learning.

Strategic model & foundations

Issuer discloses $d = \theta + m$ at cost $m^2/2\gamma$. The rule posts a public score; investors bid; allocation determines proceeds k ; the outcome $y = \theta + pk + \varepsilon$ then enters later calibration.
Issuer types differ in fundamental quality θ , private value of capital v , and social marginal product p . Under quadratic cost and single crossing, the PBE benchmark gives manipulation increasing in perceived evaluator sensitivity.

Three complementary lenses

Game theory: strategic disclosure increases with the evaluator's perceived sensitivity; equilibrium can anticipate average inflation, so the cost can appear as resources spent optimizing for the machine.
Social choice: pricing accuracy, allocative efficiency, and retail access can conflict; later issuers inherit the calibration externality.
Mechanism design: the allocation rule affects whether the evaluator's feedback footprint can be identified.

Auction or allocation comparison

100 IPO shares; 15 institutional and 35 retail investors. Investors observe the public AI signal and private signals.
M1 uniform-price: bids sorted until shares are exhausted; winners pay the lowest accepted price.
M2 bookbuilding: underwriter discretion; the model assigns 85% to institutions and applies an 8% pricing discount.
Boundary: memory is cross-firm; the model abstracts from seasoned repeat issuance.

02 Design & evidence

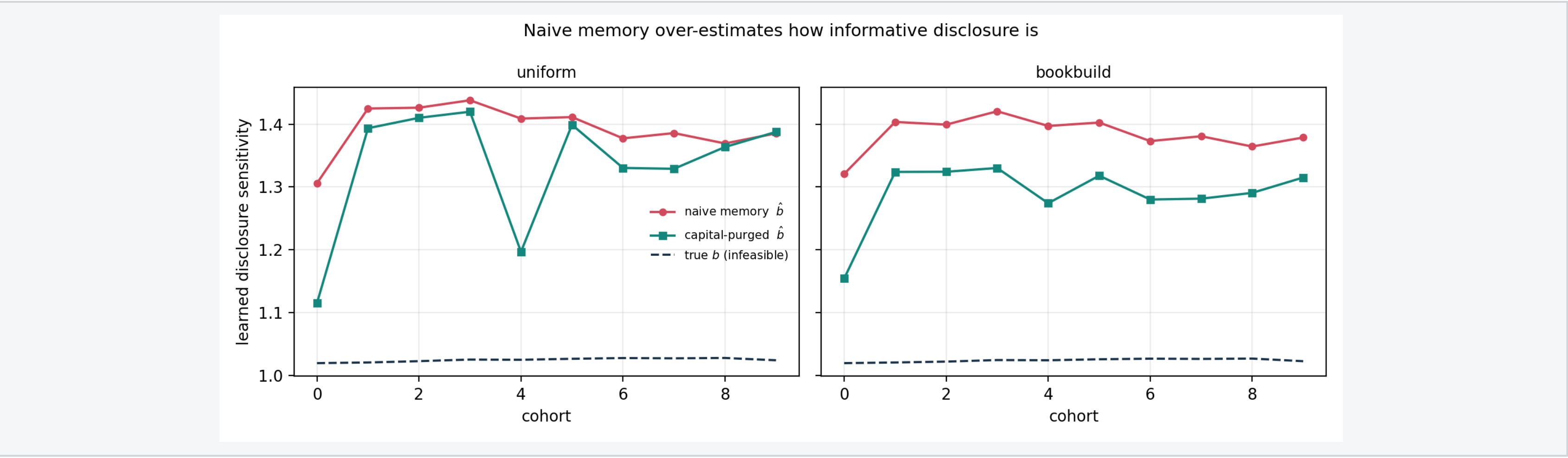


Fig 1. What the rule learns, cohort by cohort, against the infeasible truth.

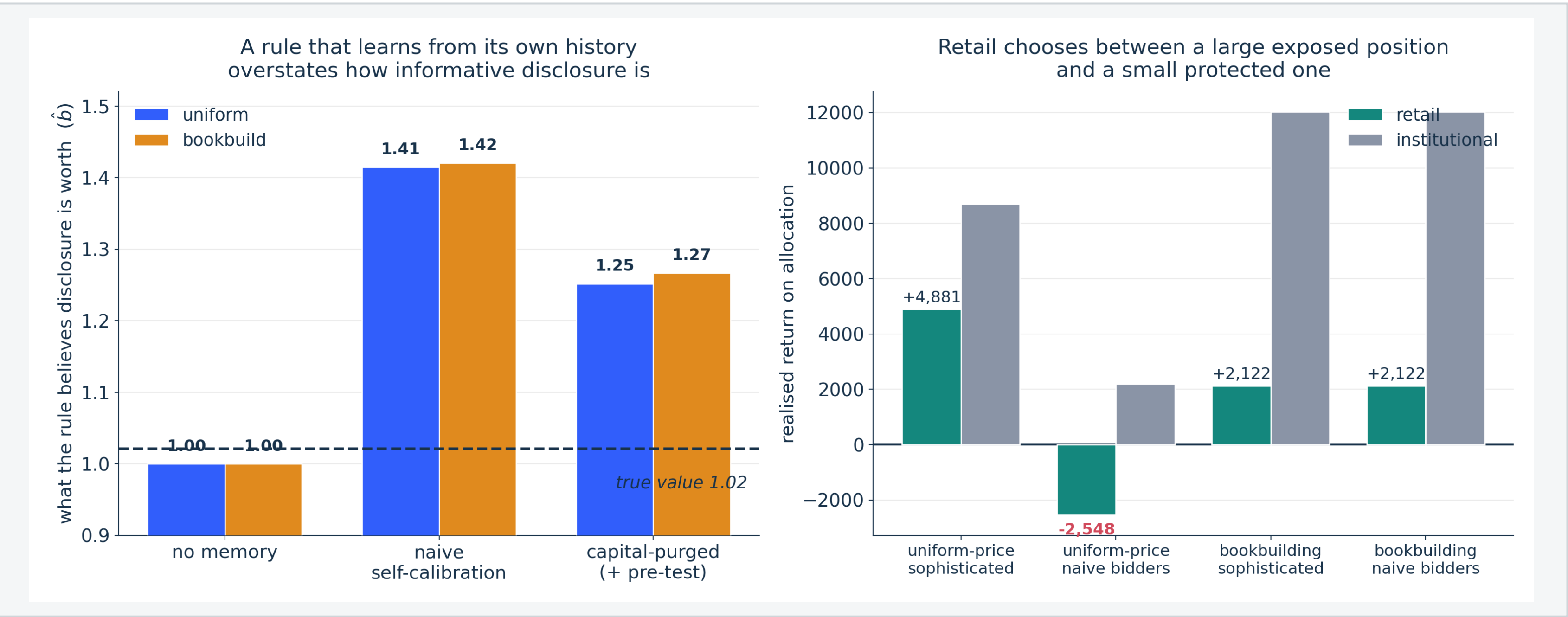


Fig 2. 200 issuers, seeds 42–46, two-cohort burn-in. Actual simulation output.
Naive memory raises mean manipulation from 0.408 to 1.367 (uniform-price) and 0.921 (bookbuilding). Under uniform-price, manipulation cost rises $14.305 \rightarrow 243.230$ (17 $\times$); under bookbuilding, $14.305 \rightarrow 91.928$ (6.4 $\times$). Capital purging reduces feedback bias under uniform-price ($5.344 \rightarrow 4.894$; $F = 0.136$) but not under bookbuilding ($2.012 \rightarrow 2.527$; $F = 0.900$). With no purge, price RMSE rises $0.551 \rightarrow 30.328$ under uniform-price.
Unresolved: a residual remains because p is heterogeneous and the allocation-shock first stage is weak.

03 Impact & dialogue

Practical design question

Question: when can a valuation rule safely learn from outcomes it helped create?
Potential benefit: an auditable criterion for separating genuine performance from evaluator-induced outcomes.
Risk: a naive self-calibrated model can look strong on backtests because its errors track its own past allocations.
Proposed correction: publish calibration sensitivity, the memory-update log, and identification diagnostics for outside audit.

Interdisciplinary evidence

Behavioral — Hugging Face
Exploratory test: players bid against a public AI valuation while the issuer can move the disclosed input. The planned analysis recovers each player's implied weight on the public signal.
Algorithm appreciation and anchoring make competing predictions about how that weight changes when calibration status is disclosed. No completed sample or treatment estimate is claimed here; this is an exploratory direction check.
Computational — GitHub
200 synthetic issuers, 5 seeds, 3 evaluator designs $\times$ 2 allocation mechanisms. One command regenerates the reported numbers plus a run record.

Limitations & next test

Capital productivity is exogenous. The AI is a rule, not a player. One AI, no competition. All data synthetic.
Load-bearing: exogeneity of the allocation shock. True by construction here; in real data rationing may track unobserved demand. Our identification is a PROPOSAL.
Next test: robustness to allocation-shock assumptions, then pre-registered replication on real issuance data.

SEP 28 Symposium, peer evaluation & case impact
SEP 30 Final update

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Selected references

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